

ITS 472 Network Design – Project Report

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i. Team Responsibilities

Dominic Leon: IP allocation, BGP, OSPF, ACL

Marcus Nuñez: Topology, RIPv1/2, VLANs, web/FTP servers, report

I. Introduction & Topology

This project implements a multi-autonomous system (AS) network consisting of three separate domains, designated as AS10, AS20, and AS30. Each domain utilizes their own Interior Gateway Protocol (IGP) for general AS communication, Border Gateway Protocol (BGP) for inter-AS communication, as well as two configured VLANs separating the devices in AS20. Access Control Lists (ACL) are also implemented, giving access authority to devices in AS10 that only allows outward communication to other autonomous systems. The routers in this network make use of multiple routing protocols to communicate with one another, including Open Shortest Path First (OSPF), Routing Information Protocol (RIP) versions 1 and 2, and Internal/External Gateway Protocol (IGP/EGP). The network topology consists of four PCs, two web servers, one FTP server, and four switches, and four routers. These devices are grouped accordingly into the three autonomous systems, following the provided project design. The physical design requires a total of 12 Ethernet cables and two serial cables, but the use of virtual machines to host the server devices reduced this amount. Using this method, PC1 hosts FTP1, PC4 hosts WEB2, and Marcus' personal laptop hosts WEB1. All three servers are ran in Ubuntu through VirtualBox.

IP addressing utilized subnetting in order to efficiently divide networks to meet host requirements, including extra space in case of later growth. The larger subnet masks of /24 were chosen to allow for the growth of end devices, /29 being chosen to limit the amount of addresses just above the requirements for all for routers, reducing the waste of addresses. VLAN10 utilizes a mask of /25 to allow for requirement of 120 devices with six addresses to spare.

AS10:

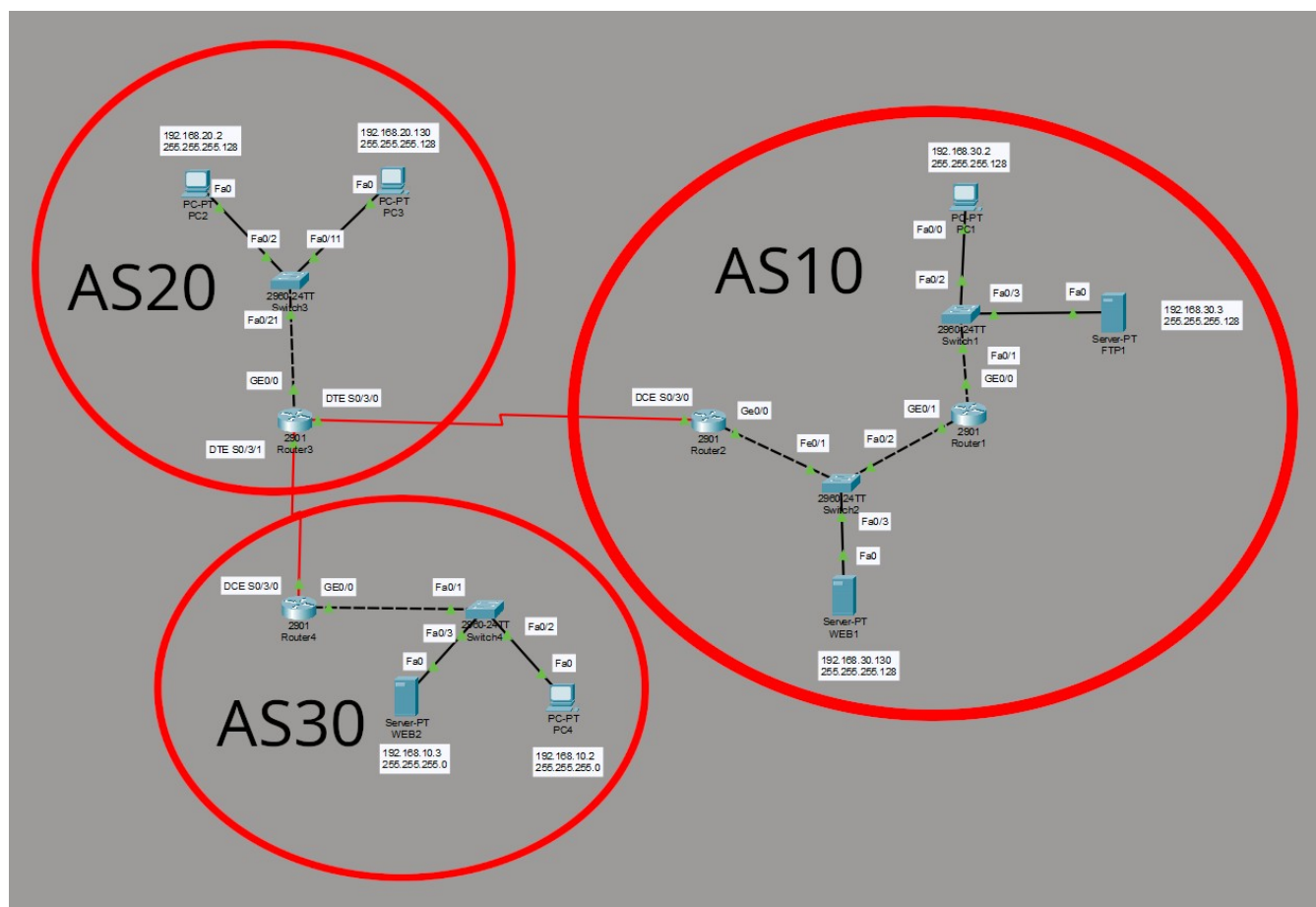
Device / Interface	IP Address	Subnet Mask	Default Gateway
PC1	192.168.30.2	/25 255.255.255.128	192.168.30.1
FTP1	192.168.30.3	/25 255.255.255.128	192.168.30.1
WEB1	192.168.30.130	/25 255.255.255.128	192.168.30.129
R1 g0/0 (to S1)	192.168.30.1	/25 255.255.255.128	
R1 g0/1 (to S2)	192.168.30.129	/25 255.255.255.128	
R2 g0/0 (to S2)	192.168.30.131	/25 255.255.255.128	
R2 s0/3/0 (to R3)	10.0.0.1	/30 255.255.255.248	

AS20:

Device / Interface	IP Address	Subnet Mask	Default Gateway
PC2 (VLAN10)	192.168.20.2	/25 255.255.255.128	192.168.20.1
PC3 (VLAN20)	192.168.20.130	/25 255.255.255.128	192.168.20.129
R3 s0/3/0 (to R2)	10.0.0.2	/30 255.255.255.252	
R3 s0/3/1 (to R4)	10.0.1.1	/30 255.255.255.252	
R3 g0/0 (to S3)	192.168.20.1	/25 255.255.255.128	

AS30:

Device / Interface	IP Address	Subnet Mask	Default Gateway
PC4	192.168.10.2	/24 255.255.255.0	192.168.10.1
WEB2	192.168.10.3	/24 255.255.255.0	192.168.10.1
R4 s0/3/0 (to R3)	10.0.1.2	/30 255.255.255.252	
R4 g0/0 (to S4)	192.168.10.1	/24 255.255.255.0	



II. BGP & IGP Configuration

Border Gateway Protocol is implemented as the Exterior Gateway Protocol, allowing for communication between the three autonomous systems while maintaining internal independent routing policies. In AS10, R2 is configured with a process number of 10, R3's neighbor status configured at its serial IP of 10.0.0.2. R2 then advertises the addresses 192.168.30.0 /24 and 10.0.0.2 for AS10, allowing the other routers to learn where to direct traffic from those associated IP addresses. AS30 follows a fairly similar BGP assignment, being configured with a process number of 30, establishing R4's neighbor relationship with R3 as 10.0.1.1, and advertising its internal networks as 192.168.10.0/24. In AS20, R3 serves as a transit router between AS10 and AS30, configured with a process number of 20. R3's required neighbor relationship of R2 (10.0.0.2) and R4 (10.0.1.2), in combination with its advertising of both VLAN addresses (192.168.20.0/25 and 192.168.20.128/25), make it an essential component for all external traffic.

BGP summary for R2, R3, and R4:

```
R2#show ip bgp
BGP table version is 14, local router ID is 192.168.30.131
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 192.168.10.0    10.0.0.2                0         0 20 30 i
*> 192.168.20.0/25 10.0.0.2                0         0 20 i
*> 192.168.20.128/25
   10.0.0.2                0         0 20 i
*> 192.168.30.0/25 192.168.30.129          2        32768 i
*> 192.168.30.128/25
   0.0.0.0                0        32768 i

R3#show ip bgp
BGP table version is 18, local router ID is 192.168.20.129
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 192.168.10.0    10.0.1.2                0         0 30 i
*> 192.168.20.0/25 0.0.0.0                0        32768 i
*> 192.168.20.128/25
   0.0.0.0                0        32768 i
*> 192.168.30.0/25 10.0.0.1                2         0 10 i
*> 192.168.30.128/25
   10.0.0.1                0         0 10 i

R4#show ip bgp
BGP table version is 6, local router ID is 192.168.10.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 192.168.10.0    0.0.0.0                0        32768 i
*> 192.168.20.0/25 10.0.1.1                0         0 20 i
*> 192.168.20.128/25
   10.0.1.1                0         0 20 i
*> 192.168.30.0/25 10.0.1.1                0         0 20 10 i
*> 192.168.30.128/25
   10.0.1.1                0         0 20 10 i
```

Each autonomous system makes use of a different IGP method, separating traffic while ensuring they can still communicate between one another. R1 and R2 of AS10 use OSPF with a process ID of 1

in area 0, advertising the addresses 192.168.30.0/25 and 192.168.30.128/25 respectively with a point-to-point configuration. AS20 uses RIP version 2 with auto-summary disabled for clearer traffic direction. R3 advertises LAN gateway address, 192.168.20.0, as well as the serial addresses connecting to routers R2 and R4, 10.0.0.0 and 10.0.1.0, under passive interfaces to reduce unnecessary updates. AS30 uses RIP version 1, simply advertising its input, 10.0.1.0, and its output to S4, 192.168.10.0.

R1 and R2 OSPF summary:

```
R1#show ip ospf database

      OSPF Router with ID (192.168.30.129) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum Link count
192.168.30.129 192.168.30.129 785          0x80000009    0x001B38 3
192.168.30.131 192.168.30.131 817          0x80000008    0x00CE41 3

      Type-5 AS External Link States

Link ID        ADV Router    Age          Seq#           Checksum Tag
192.168.10.0   192.168.30.131 315          0x80000002    0x00F8A7 20
192.168.20.0   192.168.30.131 817          0x80000002    0x008D88 20
192.168.20.128 192.168.30.131 817          0x80000002    0x00880D 20

R2#show ip ospf database

      OSPF Router with ID (192.168.30.131) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#           Checksum Link count
192.168.30.129 192.168.30.129 730          0x80000009    0x001B38 3
192.168.30.131 192.168.30.131 760          0x80000008    0x00CE41 3

      Type-5 AS External Link States

Link ID        ADV Router    Age          Seq#           Checksum Tag
192.168.10.0   192.168.30.131 258          0x80000002    0x00F8A7 20
192.168.20.0   192.168.30.131 760          0x80000002    0x008D88 20
192.168.20.128 192.168.30.131 760          0x80000002    0x00880D 20
```

R3 and R4 RIP summary:

```
R3#show ip rip database
10.0.0.0/8      auto-summary
10.0.0.0/30     directly connected, Serial0/0/0
10.0.1.0/30     directly connected, Serial0/0/1
192.168.20.0/24 auto-summary
192.168.20.0/25 directly connected, GigabitEthernet0/0.10
192.168.20.128/25 directly connected, GigabitEthernet0/0.20

R4#show ip rip database
10.0.0.0/8      auto-summary
10.0.0.0/30
    [1] via 10.0.1.1, 00:00:22, Serial0/0/0
10.0.1.0/30     directly connected, Serial0/0/0
192.168.10.0/24 auto-summary
192.168.10.0/24 directly connected, GigabitEthernet0/0
```

III. VLAN & ACL Configuration

In order to split AS20's devices into two separate VLANs, configuration was implemented on its border switch, S3. Both VLANs were each assigned 10 Ethernet ports, VLAN10 receiving fa0/1-10, where PC2 resides, and VLAN20 receiving fa0/11-20, where PC3 is connected. R3 is connected through port fa0/21, configured with a trunk link using 802.1Q encapsulation, allowing both VLANs to communicate with the same router. On R3, router-on-a-stick was configured to view each VLAN as sub-interfaces G0/0.10 at address 192.168.20.1/25 and G0/0.20 address 192.168.20.129/25 respectively, allowing for inter-communication between VLANs.

R3 IP Route:

```
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Prot
GigabitEthernet0/0	unassigned	YES	NVRAM	up	up
GigabitEthernet0/0.10	192.168.20.1	YES	NVRAM	up	up
GigabitEthernet0/0.20	192.168.20.129	YES	NVRAM	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
Serial0/0/0	10.0.0.2	YES	NVRAM	up	up
Serial0/0/1	10.0.1.1	YES	NVRAM	up	up

S3 VLAN Summary:

```
S3#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/22, Fa0/23, Fa0/24, Gi0/1 Gi0/2
10 VLAN10	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
20 VLAN20	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

R1 hosts ACL configuration in order to meet project required restrictions, most notably the one-way traffic from AS10 while still allowing access to WEB1. Access group 101 is used to restrict FTP

access on ports 20 and 21, only allowing internal access from PC1 to FTP1. Access group 102 controls ping traffic to PC1, only allowing outbound communications, except for when responding to PC1.

R1 Access-Lists:

```
R1#show access-list
Extended IP access list 101
 10 permit tcp host 192.168.30.2 host 192.168.30.3 eq ftp
 20 permit tcp host 192.168.30.2 host 192.168.30.3 eq ftp-data
 30 deny tcp any host 192.168.30.3 eq ftp
 40 deny tcp any host 192.168.30.3 eq ftp-data
 50 permit ip any any (5 matches)
Extended IP access list 102
 10 permit ospf any any (349 matches)
 20 deny icmp 192.168.20.0 0.0.0.255 host 192.168.30.2 echo
 30 deny icmp 192.168.10.0 0.0.0.255 host 192.168.30.2 echo
 40 permit icmp any any echo-reply
 50 permit ip any any (941 matches)
```

IV. Conclusion

The completed network satisfies indicated project requirements to our best abilities, devices in AS20 and AS30 (PC2, PC3, and PC4) being able to successfully ping one another, as well as ACL restrictions only allowing outbound communication from PC1. Both web servers have also been configured for access by all PCs, the FTP server's access only being limited to PC1, and AS20 configuration properly assigns devices their respective VLANs with 10 ports each and working intercommunication. The use of BGP enables communication between independent autonomous systems, while OSPF and RIP provide efficient internal routing within each domain. Overall, we believe our project successfully demonstrates the implementation of a multi-autonomous system network, utilizing various routing protocols, VLAN segmentation, and access control mechanisms.